



Introduction to Jenkins

A comprehensive overview of the leading open-source automation server powering modern DevOps workflows

What is Jenkins?

Jenkins is an open-source automation server that enables developers to build, test, and deploy their software reliably through continuous integration and continuous delivery (CI/CD) practices.



Automation Server

Jenkins automates the software development processes including building, testing, and deployment, reducing manual intervention.



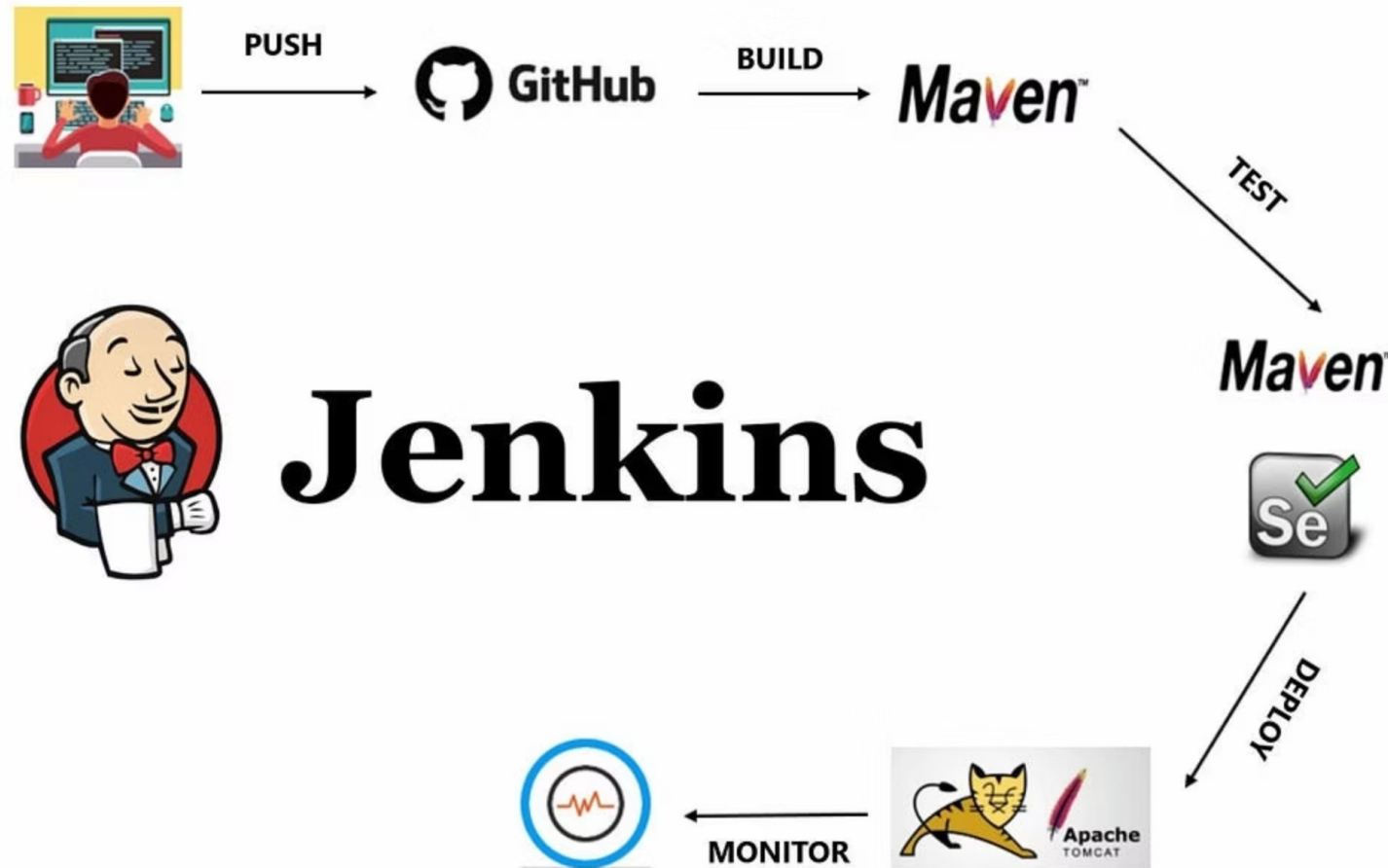
CI/CD Enabler

Facilitates continuous integration by automatically testing code changes and continuous delivery by deploying validated code to production.



Extensible Platform

Features a robust plugin ecosystem with over 1,800 plugins that extend its functionality to integrate with virtually any tool in the development lifecycle.



History and Evolution

- 1** — **2004**
Hudson project started at Sun Microsystems
- 2** — **2011**
Forked as Jenkins after Oracle acquisition
- 3** — **2016**
Jenkins 2.0 released with Pipeline as code
- 4** — **2018+**
Cloud-native features and Kubernetes integration

Why Jenkins is Popular

Jenkins has maintained its position as a leading CI/CD tool in the DevOps ecosystem for several compelling reasons:

Open Source & Free

Zero licensing costs with enterprise-grade capabilities and community support

Extensive Plugin Ecosystem

Over 1,800 plugins enabling integration with virtually any development tool

Flexibility & Customization

Adaptable to diverse workflows and technology stacks with Pipeline as Code

Strong Community

Active development, regular updates, and extensive documentation

These factors have made Jenkins the tool of choice for organizations ranging from small startups to large enterprises implementing DevOps practices.

Jenkins vs. Other CI/CD Tools

Tool	Strengths	Limitations
Jenkins	Highly customizable, extensive plugins, mature ecosystem	Requires maintenance, server management
GitHub Actions	Tight GitHub integration, cloud-hosted, YAML-based	Limited to GitHub, newer ecosystem
GitLab CI	Integrated with GitLab, container-native, YAML-based	Best with GitLab repositories
Azure DevOps	Microsoft ecosystem integration, comprehensive ALM	Less flexible, commercial pricing

While newer tools offer cloud-native approaches with less maintenance overhead, Jenkins remains popular for its unmatched flexibility and extensibility, especially in complex enterprise environments with diverse technology stacks.

Real-World Use Cases



Automated Testing

Netflix uses Jenkins to run thousands of automated tests daily, ensuring high-quality streaming experiences across devices.



Continuous Deployment

Amazon leverages Jenkins for continuous deployment pipelines that push code changes to production multiple times per day.



Security Scanning

Financial institutions implement Jenkins pipelines with security scanning tools to identify vulnerabilities before deployment.

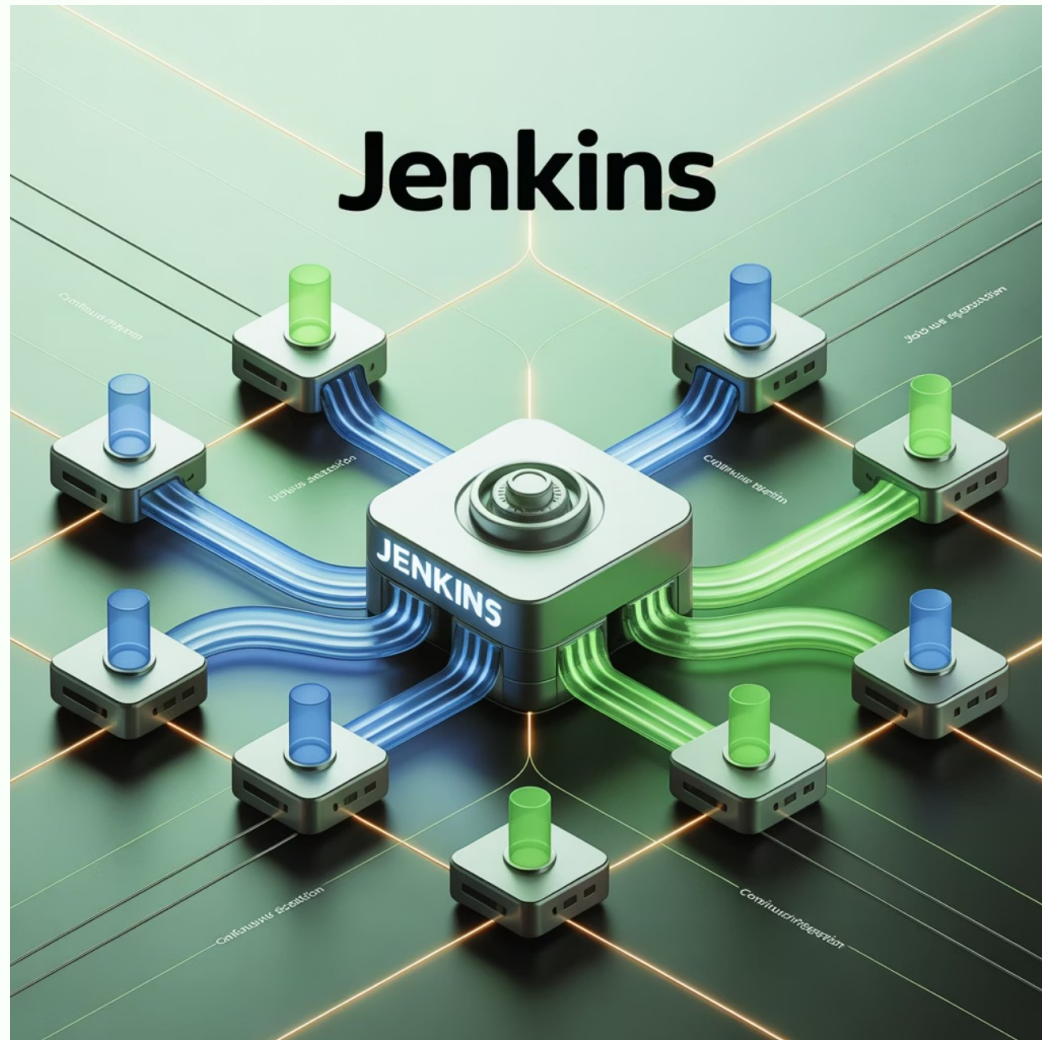


Mobile App Builds

Mobile app developers use Jenkins to automate builds for iOS and Android platforms, managing complex signing and distribution processes.

These examples demonstrate Jenkins' versatility across industries and use cases, from startups to Fortune 500 companies.

Key Jenkins Components



1 Jenkins Master

Central controller that schedules jobs, distributes work to agents, and provides the web UI for configuration and monitoring.

2 Jenkins Agents

Worker nodes that execute jobs in distributed environments, enabling parallel execution and heterogeneous build environments.

3 Pipelines

Code-defined workflows that specify the entire CI/CD process, enabling version control and treating infrastructure as code.

4 Plugins

Extensions that add functionality and integrations with other tools in the development ecosystem.

Getting Started with Jenkins

Installation

Install Jenkins on Linux, Windows, macOS, or deploy via Docker containers and Kubernetes.

```
docker run -p 8080:8080 -p  
50000:50000 jenkins/jenkins:lts
```

Initial Configuration

Set up security, install essential plugins, and configure your first pipeline using the intuitive web interface.

Create Your First Pipeline

Define your CI/CD workflow as code using the Jenkinsfile format, either through the UI or directly in your repository.

Resources: [Official Documentation](#) | [Plugin Index](#) | [Pipeline Examples](#)

Thank you for your attention! Questions?